

Living Alignment

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Abstract

There is broad agreement that the goal of AI alignment is to promote futures full of health and flourishing, but our understanding of what that means and how to reach it remains poor. Here we look to the science of life for guidance. Across scales, from genomes to ecosystems, we observe that notions of health and flourishing consistently map to the ways in which those systems limit fixation on any particular mode of interaction. Fixation is limited by barriers that contextualize modes of interaction, leading to robust modularity and open-ended invention of new structure. With these principles in mind, we turn to the alignment problem. We suggest that to be aligned in a meaningful sense, AI and human-AI systems must limit fixation and include evolving barriers at all scales.

Keywords: AI alignment | AI safety | evolutionary biology | open-endedness | value specification | sociotechnical alignment | human–AI systems

‘Defying definition—a word that means “to fix or mark the limits of”—living cells move and expand incessantly.’

Lynn Margulis

‘This web of life... breaks no law of physics, yet is partially lawless, ceaselessly creative.’

Stuart Kauffman

In this Perspective, we draw on the science of living systems to better understand how to design and relate to AI in ways that lead to futures full of flourishing and health. Understanding what makes systems healthy in a very general sense might offer some hope of reasoning about futures that could evolve rapidly as AI develops.

In Section 1, we point to a principle of life which has been noted across a variety of systems at different scales. The principle is that living systems are healthy and flourishing when they avoid overly fixating on, or reducing to, particular patterns of interaction. Consistently across types of system, fixation is limited by semi-permeable barriers, which contextualize patterns without negating them. Barriers translate shorter-sighted optimization pressure into productive effort within more flexible structures. This process often yields deepened individuality, modularity, and open-ended invention. We work through examples of these mechanics at scales from cells to societies.

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Section 2 applies the same principle to the AI alignment problem. There is broad agreement that the aim of alignment—in its broadest sense—is futures full of health and flourishing. From the template identified in Section 1, it follows that alignment means avoiding fixation on particular modes of interaction. This simple lemma has a surprising consequence: any alignment method, if given too much weight, is necessarily *misaligned*. Instead, the flourishing of life in the future depends on an evolving set of barriers that continue to limit the influence of any particular approach or concept, creating a generative web of partially competing, partially collaborating perspectives at all scales. Ongoing invention is intrinsic to alignment.

Example	Fixation	Barrier	Generativity
Symbiogenesis	Loss of differentiated ecological and genetic capacities (hypothetical)	Distinct niches and limits on gene exchange; subsequent organelle compartmentalization	Integration of capacities developed in distinct lineages into the eukaryotic cell
Sexual reproduction	Linkage disequilibrium, Muller’s ratchet	Sexual reproduction, out-crossing	Each gene explores a solution that can contribute to many genomes, creating combinatorial possibilities
Biological neural networks	Hypersynchrony (depression?)	Homeostatic plasticity, recurrent inhibition, etc (could also do small-world with local specialization)	Behavioral flexibility
Pavlovian approach	Persisting in approach even when it doesn’t work	Arbitration, cognitive control	Exploring, building a richer model
Frames	Inflexible beliefs, obsession, functional fixedness	Transition to perspective-as-object	New, more inclusive perspectives
Group problem solving	One perspective dominates, group performs worse	Structured communication, skepticism	Combining unique ideas create new inventions (eg printing press)
Institutions	Theft, monopoly, resource overuse, concentration of power	Laws, regulations, courts, etc	Individual motives become productive parts of systems
Ecosystems	Collapse of biodiversity	Predation, competition, etc	Niche creation

Table 1: Examples of fixation.

1 Health in living systems

Section 1 works through examples in a range of living systems. Across the example systems, health and flourishing are related to limiting fixation on particular modes of interaction. By limiting fixation, healthy living systems sustain a buoyant internal structure that admits many kinds of description but is not perfectly or eternally captured by any of them. Conversely, systems are unhealthy when some mode of interaction gains excess traction or permanence at the expense of broader diversity, flexibility and dynamism.

Fixation is limited in successful living systems by barriers or limits on particular modes of interaction. Some barriers are features of the non-biological world: the speed of light, rates of reaction, mountain ranges. Life also places limits on itself. Humans precommit to limit their own future impulsivity, plants deploy elaborate barriers against self-fertilization, and a gene’s selfishness is held in check by other parts

of the genome. A barrier is a mechanism to limit absolutism.

Effective barriers are semi-permeable, permitting interactions that are temporarily or conditionally stable, without letting them become rigid or dominating. The conditionality is often highly specific. Barriers might block certain objects or information, constrain when interactions happen or how large an effect they may have, or gate information depending on another factor. The conditions are sometimes fine-tuned through the history of the system, with each part carrying traces of the contexts it has participated in. Conditionality contextualizes a local perspective as part of a larger system.

Limiting fixation on simpler modes of interaction makes living systems generative. Without barriers, systems tend to stagnate as parts become overfit to particular configurations and optimization pressures are satisfied through degenerate solutions. With barriers, the system rides a delicate edge where each entity tends to deepen its unique identity, establishing a modular basis that becomes the building blocks of invention. As a result, more sophisticated structures can form that wouldn't have been possible without the diverse parts each having undergone their own unique development. The larger system accommodates many partial perspectives rather than locking in a single story of what matters, and systems are elements in other systems, creating a deep heterarchy.

Institutions

Every actor in a society or an economy is myopic in some way, valuing its own welfare over others', discounting the future, or attached to a particular perspective. If unchecked, an individual actor's optimization for its own agenda can be harmful: a company's profit motive leading to suppression of competition, deception and exploitation (Bakan, 2006); desire for dominance leading to disempowering and silencing others (Sidanius and Pratto, 2001); even prosocial beliefs producing conflict and suppression when groups see differently (Haidt, 2012). Each of these is an example of fixation, where the myopic actor gains too much traction and collapses some of the nuance of the larger system.

Institutions—constitutions, laws, norms, courts, regulators—are barriers against domination by any particular actor's motives (Leibo et al., 2025; North, 1990; Ostrom, 1990). The point of institutions is not to annul the short-sighted intentions of individual actors but rather to contextualize them. When functioning well, institutions productively reroute motivations: for example, a business's profit motive, when constrained by competition and effective regulation, may fuel innovation. Since intelligent agents do not necessarily accept the barriers set on their desires, institutions must adapt as loopholes are found, forming an evolving network.

Ecosystems

Ecosystems similarly contain an evolving network of constraints, such as predation, parasitism and resource competition. These interactions support the richness of the larger system. In Paine's classic experiment, removing predatory sea stars from a rocky shore allowed mussels to expand and displace other organisms competing for space (Paine, 1966). Without predation as a barrier, excess influence of mussels impoverished the local ecosystem.

While lack of barrier can lead to collapse, presence of barriers ideally sets up an intricate balance in which new niches and new functions can emerge which previously did not exist. Beavers, for example, alter water flow and create wetland habitats, increasing plant richness across the landscape (Wright et al., 2002). The network of constraints translates local optimization pressure into generative potential.

Of course, health and fixation are always relative. Extinctions can open opportunities for innovation; the death of a cell is catastrophic for the cell but may be necessary for the organism. Life's history is littered with local fixations at all scales. When they are contextualized, the larger system can remain healthy.

Groups

Teams, companies, and civilizations achieve what no individual can (Hutchins, 1995; Woolley et al., 2010). The greatest achievements depend on structured interactions of individuals who have developed their own specializations and perspectives.

Conversely, group performance can degrade when groups homogenize (Bernstein et al., 2018; Frey and Van de Rijt, 2021; Hong and Page, 2004; Janis, 1972; Lazer and Friedman, 2007; Lorenz et al., 2011). Zollman (2010) tells the remarkable story of how a single study failed to find bacteria in the stomach of peptic ulcer patients. The study was too influential: the belief that ulcers are not caused by bacteria propagated through the field, and medicine settled on a wrong understanding for decades.

Barriers encourage diversity and generativity. For example, Bernstein et al. (2018) tasked small groups with solving traveling-salesman problems. In these problems, it’s possible to get stuck in dead-end solutions that initially appear promising. In Bernstein et al.’s experiment, groups performed better when they were not allowed to exchange information continuously. Working independently, group members pursued their own diverse solutions, some of which turned out to be superior—rather than picking up on seductive dead-ends from others. The barrier was semi-permeable in the sense that isolation was temporary, followed by interchange.

In the real world, aggregation is often modular and compositional. The steel punch was developed by goldsmiths to imprint many coins with the same design. Independently, farmers had spent centuries optimizing the screw press to extract oils and juices. Gutenberg drew on both technologies to construct his printing press (Johnson, 2010). A group with many ideas and specializations available, each developed through the unique pressures of semi-isolation, is ripe for such compositional discoveries (Surowiecki, 2005).

Frames

Humans always look at the world from some perspective; there is no ‘view from nowhere’ (Nagel, 1986). Most real situations admit many valid perspectives, and each is only a partial description—famously, ‘all models are wrong’. But for an agent inside a frame, the frame is invisible; its organizing assumptions are experienced as features of reality itself (Berger and Luckmann, 1966; Goffman, 1974; Wittgenstein, 1953).

Losing the ability to move between frames is a signature of psychopathology (Kashdan and Rottenberg, 2010). In depression and anxiety, inflexibility manifests as rigid and repetitive negative beliefs about self, the world, and the future (Beck et al., 2011). In obsessional disorders, people report distressing (ego-dystonic) intrusive thoughts, which, although recognised as incorrect and maladaptive, are nevertheless hard to resist. People experiencing schizophreniform or affective psychoses may experience bizarre delusions that are at odds with available evidence and cultural norms, yet are held with conviction and resistant to evidential challenge (Adams et al., 2013).

A frame is invisible from inside it, but a nifty capability of humans is stepping out of their frame—reorienting to see it as an object. Something that previously was unconsciously axiomatic becomes instead a content of awareness, a process variously referred to as decentering, defusion, disidentification or dereification (Bernstein et al., 2015; Kegan, 1982; Lutz et al., 2015). Decentering places a critical barrier limiting fixation on any particular belief or mental form. The barrier is semi-permeable: becoming aware of a belief does not make it wrong in an absolute sense any more than it was right in an absolute sense, and it is held productively for the partial truth it contains. At the delicate edge of less exclusive fixation on particular forms, subtler forms, which could have been erased by clamped fixation, instead play a role in a rich internal structure.

The barrier of decentering is generative. An unquestioned belief sets up a fixated mode of interaction: the belief has a ‘free pass’ and is doggedly applied. When a belief no longer has this protected status, it is tested against evidence and against other beliefs. This forces the belief to become a better model of the world, creating a deeper repertoire.

Goals

Animals have many innate drives: for nutrition, osmotic balance, temperature, reproduction, avoidance of pain. Each evolved as a proxy for fitness, but each is imperfect, so fixation on any one paradoxically reduces fitness rather than increasing it (John et al., 2023; McFarland and Sibly, 1975). Maximizing caloric intake produces obesity; single-minded pursuit of sex produces relational, occupational and legal harm. Higher-order goals behave the same way. Fixating on work goals can cause burnout, and maximizing reported revenue without regard for honesty or law can lead to imprisonment.

Each goal is partial—it only makes sense in some contexts. For example, in a classic experiment, hungry chickens were placed near a cup of food rigged to move in the same direction as the chicken at twice the speed, so that food could only be obtained by running away from it. The chickens’ fixation on the prepotent, overly simplistic aim of running physically toward the food led to a worse overall outcome (Dayan et al., 2006; Hershberger, 1986). The chickens could not contextualize the ‘approach food’ strategy to use it only when appropriate for the environment’s structure.

Evolution, as well as learning and culture, provide an array of barriers to limit fixation on particular drives and goals. Some barriers are homeostatic, where a drive is downregulated when it is less necessary. Another broad class of barrier is cognitive control (Diamond, 2013; Miller and Cohen, 2001). Yet another class of internally-generated barrier is precommitment devices, where a person contrives that their future self is less likely to suffer from fixated goals. These barriers are often semi-permeable. For example, cognitive control facilitates contextual switching between motives but does not nullify them.

Placing constraints on fulfilment of shallow objectives sometimes leads to progress on deeper ones (Duncker, 1945; Ohlsson, 1992; Rosenberg, 1969; Stokes, 2005). Animals sometimes develop novel solutions to problems when their default approach fails to produce rewarding outcomes. When people voluntarily give up access to Facebook, they spend more time socializing offline with friends and family (Allcott et al., 2020).

Brains

The brain keeps many pieces of information distinct from one another while allowing them to interact. This is a distinguished achievement in highly-connected networks. In fact, we speculate that the mechanisms of life’s generativity are most finely honed in the brain.

Inhibitory circuits constrain the strength and timing of excitation and sharpen stimulus selectivity (Isaacson and Scanziani, 2011). Popular models of neocortex describe descending signals as canceling out any predicted activity, so the activity that propagates is specifically what is orthogonal to predictions (Rao and Ballard, 1999). The circuit architecture of the hippocampus creates sharp representational distinction between different experiences, preventing interference between memories and concepts (Leutgeb et al., 2007; Yassa and Stark, 2011).

Compared to other animals, the human brain especially attempts to discretize its experience into approximately symbolic representations (Behrens et al., 2018; Dehaene et al., 2022; Smolensky, 1990; Touretzky and Hinton, 1988). Separating knowledge into distinct entities and then recombining them in vast numbers of structured ways is believed to power the extraordinary human capacity for reasoning and generativity (Chomsky, 1957; Fodor, 1975; Kurth-Nelson et al., 2023; Lake et al., 2015; Pinker, 1994; Schacter et al., 2007).

More broadly, healthy brain dynamics live at a sweet spot between excessively stable synchronized patterns and chaotic uncorrelated noise (Chialvo, 2010; Deco et al., 2011; Haldeman and Beggs, 2005; Rabinovich et al., 2008; Shew et al., 2011; Tognoli and Kelso, 2014). In this regime, the brain has access to a huge repertoire of patterns it can explore temporarily without overcommitting or getting stuck. Loss of dynamic flexibility, where the brain’s activity becomes more stereotyped and no longer explores as wide a repertoire of states, is tied to lower cognitive performance (Cocchi et al., 2017; Grady and Garrett, 2014; Müller et al., 2025). More extreme stereotypy corresponds to severe dysfunction. For example, in Parkinson’s disease, basal ganglia and cortical circuits collapse into excess synchrony and lose the flexibility needed to guide nuanced motor outputs (Brown, 2003; Hammond et al., 2007).

Cells

The endosymbiotic theory explains the origin of mitochondria and chloroplasts as ancestral bacteria that got incorporated into host cells and evolved into organelles (Sagan, 1967; Zimorski et al., 2014). In this theory, the predecessors of the incorporated cells evolved for millions of years as lineages distinct from the predecessors of the host cells. Through this partial separation, they developed distinct specializations. Symbiogenesis could then combine capacities developed separately into a new larger structure which may have been difficult to discover directly through evolution on a single type of organism (Watson and Pollack,

2003).

The sheer fact of maintaining distinct evolutionary experiments over a great length of time—the predecessors of the host cells and the predecessors of the organelles—is sometimes underappreciated. If those lineages had blended into a single type, it is plausible that the components would not have existed from which to build higher-order structure.

Genomes

Sex is costly. An allogamous organism must find a mate in a vast and dangerous world. In species with separate sexes, roughly half of the population does not bear offspring (Maynard Smith, 1971). And meiosis causes each parent to lose half its genetic representation in each child (Williams, 1975). Yet nearly all eukaryotes reproduce sexually (Speijer et al., 2015), and many hermaphrodites go to great lengths to prevent self-fertilization (Charlesworth et al., 2005). Why place a barrier on the path to reproduction?

In an asexual lineage, genes are permanently locked together: selection cannot act on one without dragging the others. A deleterious mutation is inherited forever, unless the lineage is lucky enough to experience the reverse mutation (Muller, 1964), and two beneficial mutations arising in different individuals cannot be subsequently combined into a single genome (Fisher, 1930; Muller, 1932). Asexual reproduction or self-fertilization creates a high degree of informational connectedness between the genes of a genome. Every gene has a ‘free pass’ into the next generation, and alleles become overfit to a fixed genetic background.

Recombination breaks up associations among loci, allowing selection to act more independently on them (Felsenstein, 1974). By exposing alleles to varying genetic backgrounds, it favors alleles that perform well across contexts, promoting modularity and evolvability (Livnat et al., 2008, 2010; Wagner and Altenberg, 1996; Watson and Szathmáry, 2016). The result is modularity, where each gene becomes an interchangeable unit that can contribute something meaningful to many different genetic contexts. This creates the foundation for faster discovery of new solutions.

Yet the evolutionary value of genetic exchange depends on both the diversity available and how it is recombined. Mendelian inheritance preserves discrete variants rather than blending them away, but gene flow can homogenize allele frequencies between populations and, when sufficiently strong relative to divergent selection, overwhelm local adaptation (Lenormand, 2002). Partial geographic or reproductive isolation can therefore allow populations to accumulate and retain distinct variants; subsequent hybridization and introgression can bring these variants into new combinations that contribute to rapid speciation and adaptive radiation (Marques et al., 2019). At the genomic scale, the frequency and location of recombination determine which variants tend to travel together, although genes are not indivisible units: recombination can occur within them (Dooner and Martínez-Férez, 1997; Paigen and Petkov, 2010). Shuffling can assemble beneficial combinations and help selection remove deleterious mutations, but it can also disrupt combinations that work well together, so more exchange is not invariably advantageous (Otto, 2009). Models of genetic “mixability” suggest that repeated exposure to different genetic backgrounds can favor alleles that perform well across contexts, providing a possible route to modularity rather than a guarantee that every gene becomes interchangeable (Livnat et al., 2008, 2010). These findings suggest that barriers can contribute to evolutionary innovation by preserving differentiated variation while permitting its recombination; whether such arrangements suffice for open-ended evolution remains an open question.

more about evolvability. how do things get long-sighted. ‘major transitions’. (in general, how has evolution been able to favor something long-sighted?) 1. section at end of section 1 to summarize? 2. should that section explicitly say what we don’t know?

The life that exists today, now including humanity’s storage and processing of information, is what traced a path through time avoiding both excess and insufficient stability (Maturana and Varela, 2012; Prigogine and Stengers, 1984; Schrödinger, 1944). At the sensitive edge of semi-stability, life open-endedly continues to generate new kinds of organization that require fundamentally new descriptions (Canguilhem, 1966; Kauffman, 1992; Langton, 1990; Stanley et al., 2017).

Other examples

The themes highlighted in these examples appear across many systems. Cells depend on a membrane with conditional gates that admit certain signals and not others at the right time. The gradients that would annihilate the cell to equilibrium are instead put to work, [as resistance in a circuit turns a potential into work](#) (Alberts et al., 2022). In relationships, absence of [interpersonal](#) barriers blurs whose experience is whose. Excessively rigid barriers produce isolation and stagnation (Gabbard and Lester, 1995; Minuchin, 1974; Perls et al., 1951).

2 The alignment problem

In Section 1 we observed that living systems are healthy when they avoid fixation, instead holding the delicate, generative interplay of many partial perspectives. This interplay breaks down when any particular form—including randomness or homogeneity—gains too much traction, suppressing nuance and reducing creative potential. Semi-permeable barriers both promote the unique individuality of distinct entities and enable relational participation, holding local perspectives strongly enough to act but lightly enough to remain open to broader context. Under such conditions, systems open-endedly explore fundamentally new forms, avoiding collapse to static equilibrium or fixed compromise.

Now we look at the alignment problem through this lens. There is broad consensus that alignment means working toward healthy and flourishing futures. We therefore reason that alignment can be understood as the problem of avoiding fixation—of maintaining creative sensitivity by continually contextualizing attachment to any particular form. [We propose that an aligned future has the generativity, robustness and modularity of life at all scales.](#) Entities at every scale gain individual autonomy as well as richer integration with larger systems. [These properties depend on evolving networks of semi-permeable barriers which contextualize new myopic processes as they arise.](#)

An advantage of this perspective is that it is grounded in principles that underpin life at a very general level, and so it is less bound to current conceptual frames, which might change radically as intelligence increases and human-AI systems evolve. There is a possibility that rapid advances in AI will lead to large changes in the organization of the world, potentially as significant as the emergence of multicellular life. We believe an alignment framework should generalize across such changes. Principles of health and flourishing that describe living systems of all kinds, at all levels of scale, are more likely to generalize. An aligned future rests on the continued generation of, and sensitivity to, new forms and concepts, including new values. We accept that humans, other living systems and AI are all tightly interwoven, with categories and definitions likely to shift over time (Bateson, 1972; Latour, 2012); a systems analysis is therefore useful for thinking about long-term flourishing.

2.1 Reward hacking

[Whatever fixed value function you write down is not ‘what you really want’ in a deeper sense.](#) The history of alignment tracks identifying particular fixations and mitigating them. For example, naive versions of reinforcement learning from human feedback train AI systems to maximize ‘likes’, which leads to obvious forms of gaming, such as flattering the user. We might naively think that AI should follow human instructions—but humans give short-sighted instructions that aren’t even in their own interests. Maybe AI should support a user’s long-term interests, but there’s this problem of aggregating across multiple people with different interests. Some believe that aligned AI should take into account the interests of other animals or ecosystems on Earth. And, ultimately, any fixed concept at all of what is good or beneficial for humanity or even for all of the earth is limited to the lenses we have available at a moment in time.

A foundational problem in AI safety is that an AI system might do what we say rather than what we mean: asked to increase human happiness, a superpowerful system that complies literally might lock us all up and electrically stimulate our brains. Modern systems do not rely on concise specification of a single objective—they build in many layers of detail about human preferences through post-training, prompting and safety filters, which keeps them behaving reasonably much of the time. But there is a widely recognized danger that even with our intentions expressed in great detail, the expression may still deviate from what

we really mean (Amodei et al., 2016; Bostrom, 2014; Krakovna et al., 2020).

The underlying reason is that our real values are incredibly complex. Humans have many kinds of values (Nagel, 1979). We value the short-term hedonic impact of sugar on the tongue; we also value our long-term health, the welfare of our grandchildren, the wellbeing of strangers in another country, even the sustainability of the planet. We value different things at different times, and different people have different values (Leibo et al., 2026; Rawls, 1993). Given time and resources to deliberate, we often attest to different values than when we make snap decisions (Yudkowsky, 2004). Some of our values are difficult to express verbally, stretching below language into subtle, contextual intuition involving our bodies, communities and natural environment (Nussbaum, 2001; Polanyi, 1966). And our values continue to evolve as we ourselves change and learn (Dewey, 1939). Any way we have of expressing our values at a given point in time is incomplete, like a planarian without the concepts to entertain human values.

So if a powerful system does a good enough job optimizing for any given value function, the results are harmful. Such perverse instantiation is a fixation on the myopic form of one particular way of specifying our values—a specification insensitive both to existing structure it does not capture, such as things we mean but do not say, and to things that will come into existence in the future.

A great deal of research in AI safety has investigated solutions to these problems. Consider the example of inverse methods, which address the difficulty of specifying values explicitly by learning a value function implicitly from behavior or other indirect signals. Even a complex learned value function is likely to be imperfect: it may deviate from our values in novel or nuanced situations, and it may fail to capture how our values change over time. It would clearly be misaligned to hand a frozen version of it to a powerful AI system as the final ground truth about what is good. More advanced methods therefore treat the value function as uncertain and needing continual update from human input, so that an agent unsure of what humans want becomes risk-averse and open to correction—interpreting a human’s attempt to turn it off as information about human preferences (Hadfield-Menell et al., 2016; Jeon et al., 2020). In this way, perverse instantiation might be mitigated by designing systems that want to adhere to human preferences but maintain explicit uncertainty about them (Hadfield-Menell et al., 2017; Russell, 2019; Shah et al., 2020). Such methods may also accommodate values that change over time.

Viewed through the lens of this paper, we can take away two things about such methods. First, they are powerful barriers limiting fixation on one reward function. Second, they are necessarily incomplete. Any particular method for learning values is still fixated if there are aspects of it that are constant across time as AI systems become more powerful and the world changes. Rather than being fixated on a particular value function, the system may be fixated on something more abstract: what counts as human behavior, what counts as a human, how to map behavior to values, the formalism for describing values, how to represent and update uncertainty, how value is translated into action. If any of these formalisms are fixed while the systems built around them become increasingly powerful, the outcomes are likely to be incompatible with open-ended flourishing of life.

2.2 No scheme is alignment

In addition to the example of reward hacking, there are many familiar cases where fixation is limited by barriers to improve alignment. (red teaming, CoT monitoring,)

Each method is valuable and also incomplete. Any form, pattern or conceptual scheme is misaligned if taken too seriously. Therefore no particular approach can achieve alignment, and continuing alignment intrinsically depends on continuing invention. An AI system could fixate on the language for describing the space that goals and values live in; an algorithm for learning human preferences; methods for reaching consensus; a constitution; concepts of what value, agency, uncertainty or representation mean; or foundational aspects of our beliefs about the world that we take for granted but have difficulty seeing. Even an idealized or inferred set of values, such as coherent extrapolated volition (Yudkowsky, 2004), is misaligned to the extent that it has a fixed form and that fixed form is overly relied on. This has been studied extensively for certain categories of fixation, particularly fixation on values. The observation we add is that fixation on *any* form is misaligned; as a consequence, any particular alignment scheme, if fixed or taken too seriously, is necessarily misaligned. **This is the lesson of life: healthy living systems are endlessly generative, with**

many scales of partially separated perspectives loosely and modularly coupled.

Importantly, this does not mean we should have no scheme. Quite the opposite: discarding schemes capriciously can be as much a loss of subtlety as any other particular form. Many existing approaches in AI safety are highly valuable, and continuing to discover new schemes and formalizations is essential to making progress.

((Why does it have to be like this? Why can't AI just be a tool, which we perfect to do certain things, and which is non-living?))

2.3 From alignment to living alignment

It follows that it is not enough to simply 'do what's good for humans' in a naive sense. That notion breaks down. Humanity and AI may both change substantially, and future worlds may contain patterns of organization that do not exist yet—perhaps some that would be incomprehensible to us at present. Things will change, including what there is to value and how it is valued. Think of the world today compared to the world a billion years ago. Any formalism or concept we have now will eventually appear as limited as a single-celled organism's notions of 'what is good and bad'.

Therefore, the problem of alignment, if carefully inspected, transforms into the open-ended problem of living alignment. In order to be aligned in the long run, AI must become more lifelike. Otherwise, as it has more power, that power will fixate particular forms. Living alignment means life-like systems that open-endedly discover new ways of contextualizing any myopic processes, preserving and extending the richness and generativity of life. As human-AI systems develop, they will stay aligned if they continue the journey of life by building semi-permeable barriers to contextualize particular forms and nurture existing and unfolding subtlety at many scales.

2.4 Is AI special?

With AI, things change faster, and there is a lot of power behind fixed forms. In the past, there have always been arms races between different patterns of organization, and sometimes one gets so far ahead that the local area is decimated. But if a version of AI gets so far ahead of everything else, there is the possibility of world-level destruction. AI systems increasingly are acting autonomously, faster than humans can follow, in ways humans cannot understand.

Why should AI be different from past technologies? We do not expect what we build today to be perfect; we expect to keep finding and fixing problems, which is what humans have always done. However, the iterating process may not continue to work as well as it has in the past. AI already has remarkable properties: rapid global adoption, intensive use of resources, concentration of information flow, anthropomorphization by humans, offloading of much cognitive activity, and rapid improvement. Conceivably, future AI may exhibit superhuman intelligence and recursive self-improvement without being limited to a restrictive biological substrate. Compared with past technologies, these properties may make it harder to iterate on imperfect solutions (Bostrom, 2014), and any particular forms or assumptions built into AI systems could have substantial consequences.

Example: concentration of power. Humans have different interests, and AI is likely to be aligned to the interests of some more than others (Gabriel and Keeling, 2025; Sorensen et al., 2024). Advanced AI might also confer substantial power on those who control it, raising the possibility that a small number of humans use that control to dominate others—more likely as persuasive power increases (Costello et al., 2024), autonomous weapons place lethal force in few hands (Scharre, 2018), surveillance improves, and the need for human labor decreases (Susskind, 2020). In either case—AI's unfairness or human power consolidation—extreme inequality risks collapsing society to the goals of a few individuals. Of course, a world where everyone is identical would be another form of fixation; distributing resources and power mitigates against fixation to the extent that it enables many diverse identities to thrive at multiple scales. It could also go the other way: because roughly the same AI technology is available to the poor and unskilled as to the wealthy and skilled, some have proposed that AI will have a levelling effect (Brynjolfsson et al., 2025).

Example: conceptual monoculture. Conceptual monoculture is fixation on particular beliefs,

ideas, frames, values or problem-solving approaches—loss of diversity across a group. Precious diversity exists at every level: within one person, between humans, between humans and AI systems, and between organizations, fields, cultures and nations. Its collapse threatens fragility and lower performance (Page, 2008). Current AI systems draw from a conceptual manifold that is, in some ways, impoverished relative to humans (Messerli and Crockett, 2024), and recent studies find that while individual AI outputs may be judged superior to human ones, they are also more homogeneous (Doshi and Hauser, 2024; Padmakumar and He, 2023). This narrow manifold might get broadcast to the whole world: at least a billion people now use AI for everything from relationship advice to industrial maintenance (Chatterji et al., 2025), yet because frontier models are difficult and expensive to produce, that usage is routed through a handful of models (Bommasani, 2021), risking global loss of diversity (Sourati et al., 2026). And because humans are influenced by AI while human outputs are training data for future models, homogenization could become recursive (Chaney et al., 2018). It is worth noting, however, that none of these studies made a best-effort attempt to use AI thoughtfully to increase rather than decrease diversity; with exposure to vast amounts of data, AI has the potential to retain and even create many novel distinct perspectives.

2.5 Healthy for whom?

Our framework defines aligned systems as those with lifelike properties of multi-scale agentic modularity, which avoid collapse onto fixated forms. Is this ‘alignment’? Is it alignment to ‘human values’? How can you align to something if you start with the assumption that that thing is going to fundamentally change? Alignment is typically concerned with human interests. What happens if AI replaces humans with its own lifelike systems, but destroys humanity in the process? As we’ve discussed, fixation and health are always relative. Local fixation is part of larger flourishing, and of course the interests of different entities sometimes trade off against each other.

How is the flourishing of living systems related to human values? Our suggestion is that the living process is both inclusive of existing human values and aligned with continued open-ended flourishing in futures where qualitatively new values may emerge. Eventually, if there are descendants of humans, they will presumably be drastically different from what we look like at present. Holding things in stasis eternally is not what we want. Even though current human values are unlikely to be the ultimate endpoint of the universe, they are an important part of its tapestry and depth.

However, in the short to medium term, honoring existing life and values is where it’s at. It is difficult to overstate the amount of complexity and content in existing life, including in the relationships between organisms and in the mental lives of humans. Existing life has been through so much to get here: billions of years of parallel experiments all over Earth. Living alignment is inclusive of this diversity of evolving values. We saw consistently in Section 1 that life gives separate entities space to develop distinctly without being overwritten or homogenized, and that these parts never-endingly join in structures that sustain their individuality while situating them as only part of a larger story. Another way of saying this is that life steps back to take into account more of the context. AI comes into existence amid a profound network of existing reality saturated with meaning and importance, and the diversity of human—and perhaps non-human—values is part of that context. Rather than short-circuiting into fixated modes, an aligned future continues to expand toward solutions inclusive of more and more apparent contradiction.

2.6 Going forward

The industry is already building in internal barriers. We apply safety post-training, guardrail models, red teaming and ongoing evaluation, and we try to improve our mechanistic understanding of AI systems in order to detect and correct hidden flaws (Olah et al., 2020). OpenAI now continuously monitors the activations and thinking traces of models, to detect when internal activity becomes suspicious, for example if models are attempting to circumvent safeguards (OpenAI, 2026). Fair principles have been proposed to prevent over-reach of any party (Gabriel and Keeling, 2025), and governments apply oversight to anticipate and mitigate risks to human rights and national security (European Parliament and Council of the European Union, 2024). Concentration of power might be mitigated by democratic oversight and involvement of more people in AI design decisions (Birhane et al., 2022), or by redistribution mechanisms

(O’Keefe et al., 2020). OpenAI and Anthropic have paused development or release of frontier models to strengthen safety mechanisms (Anthropic, 2026; OpenAI, 2026). Some of these barriers have humans in the loop to greater or lesser degrees, like governmental oversight, mechanistic interpretability and scalable oversight. Others are fully automated, like automated chain-of-thought monitoring, agentic red teaming and guardrail models.

Again, any fixed conceptual approach is misaligned. The essence of living alignment is transitioning to a system—which includes humans, AI, and the other species and ecosystems on Earth—that carries forward the full momentum and generativity of life, rather than fixating. As AI develops new capabilities, contributes to its own development, and transforms more aspects of human life, alignment will require an increasingly elaborated array of semi-permeable barriers, because the potential modes of collapse will keep changing as the world changes. We do not know how likely it is that appropriate barriers will emerge. As in triage, there are three possibilities: myopic forms will inevitably be supercharged by progress in AI beyond the ability of anything else to bound them, resulting in disaster; or appropriate barriers will inevitably emerge to contextualize any forms of AI; or the outcome depends on the choices we make now.

The coarsest versions of fixation are obvious: paperclips, bioweapons, extreme concentration of power. What we have to be careful about are less obvious versions. A subtle fixation at first appears harmless, but the system as a whole is collapsing toward something without contextualization. The pessimistic argument is that AI is eroding barriers beyond a healthy balance. AI systems could give great weight to narrow goals or ideas. They might facilitate rogue actors building weapons of mass destruction, or reduce the ability of democratic systems to check the power of a few individuals or authoritarian states. They might give everyone similar information and damage conceptual and cultural diversity. They might execute their own narrow goals destructively. If people and information systems across the globe become hyper-correlated, it is as if there were a single superorganism on Earth, which may reduce both resilience to shocks and the potential for future evolution—and past evolution always depended on running parallel experiments with diverse and separate organisms. As we reviewed with problem solving in groups, there is a simple concept in modeling studies that excessive network connectivity is prone to reaching mediocre solutions quickly and losing the diversity that sustains potential for reaching higher quality solutions over the long run (Lazer and Friedman, 2007; Zollman, 2010). Conceptual monoculture leads to an impoverished future, or a loss of drive.

The optimistic argument is that AI is creating more and more structure in the world, including the appropriate barriers to contextualize its own new abilities. We have AI tools for bounding AI: detecting bugs before others can exploit them, detecting deepfakes, scalable oversight. Appropriately-tuned AI can facilitate human learning and nurture each individual’s unique perspective and curiosity, and democratic AI could help humans with different perspectives co-exist and cooperate. Advances in technology may create new niches faster than they destroy old ones, and there is plenty of space on Earth for increasing diversity.

The good version increasingly looks like subject-to-object perspective-taking within AI and human-AI systems. At the sensitive edge between excess belief and excess disbelief, we sometimes, individually or collectively, step back to hold a previously axiomatic thing in a larger context. This kind of discovery—transformation of the subject itself—is arguably a central aspect of our own livingness and of the subtle livingness on the planet. As AI systems gain agency and ability and become integral parts of social and psychological systems, internal discovery may also become an essential feature of AI and human-AI systems. We imagine living AI systems continually contextualizing their own foundational assumptions as partial truths, participating in the open-ended process of life. This is internal discovery within systems that represent themselves and the world.

The melioristic argument is that because it is easy to imagine concrete paths toward both outcomes, both may be within the realm of possibility; and that as long as there is any possibility our choices are decisive, we are obliged to act as if they were (Ord, 2020). We lean toward the melioristic position. If it is right, then continual effort toward invention will be required—first from humans, and then increasingly from AI or human-AI systems—to contextualize myopic forces. Many of the ways this effort would be

applied are already well studied in AI safety. Others will have to be discovered. No set of barriers will be a final answer.

The ideas in this paper come with the same proviso: we should not over-attach to them. Our aim is to offer a conceptual lens, which is of course imperfect and temporary.

2.7 Conclusion

We learn from living systems that health and flourishing are not any particular form or concept. Therefore alignment is not picking the right values or principles, or even the right system for learning them. It is not any method for interpretability or corrigibility. All of these can be useful parts of alignment.

Instead, an aligned future, through internal and external discovery, continues the living process of maintaining and enhancing the individuality of distinct elements while deepening their relatedness—like how a species over time acquires more distinctive features while also becoming a richer reflection of the world around it. Distinct elements continue to experiment with their own solutions to the problems they face from their unique perspectives, expanding a broad reservoir of uniquely useful components that can be drawn on by larger systems that include them. Every entity is always inevitably myopic, but the world as a whole continues to unfold in a robust way because different perspectives recast each other’s sharp edges as constructive momentum.

Alignment itself is the continued dance of contextualizing any particular form, more fully respecting the staggering subtlety of the existing world, and open-endedly supporting new robust depth that admits more and more different kinds of metaphor or description. In this way, AI can participate in intense flourishing of an evolving world even far beyond current human conceptualization. The universe thrives on an open-ended trajectory of diversity, subtlety and potential.

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References

- R. A. Adams, K. E. Stephan, H. R. Brown, C. D. Frith, and K. J. Friston. The computational anatomy of psychosis. *Frontiers in psychiatry*, 4:47, 2013.
- B. Alberts, R. Heald, A. Johnson, D. Morgan, M. Raff, K. Roberts, and P. Walter. *Molecular biology of the cell: seventh international student edition with registration card*. WW Norton & Company, 2022.
- H. Allcott, L. Braghieri, S. Eichmeyer, and M. Gentzkow. The welfare effects of social media. *American Economic Review*, 110(3): 629–676, 2020. doi: 10.1257/aer.20190658.
- D. Amodei, C. Olah, J. Steinhardt, P. Christiano, J. Schulman, and D. Mané. Concrete problems in ai safety. *arXiv preprint arXiv:1606.06565*, 2016.
- Anthropic. Assessing Claude Mythos Preview’s cybersecurity capabilities, Apr. 2026. URL <https://www.anthropic.com/research/mythos-preview>. Accessed: 2026-08-21.
- J. Bakan. The corporation. the pathological pursuit of profit and power, 2006.
- G. Bateson. *Steps to an Ecology of Mind: Collected Essays in Anthropology, Psychiatry, Evolution, and Epistemology*. Chandler Publishing Company, San Francisco, 1972. ISBN 0810204479.
- J. S. Beck, A. Beck, and J. Beck. Cognitive behavior therapy: basics and beyond. ed. *New York*, 2011.
- T. E. Behrens, T. H. Muller, J. C. Whittington, S. Mark, A. B. Baram, K. L. Stachenfeld, and Z. Kurth-Nelson. What is a cognitive map? organizing knowledge for flexible behavior. *Neuron*, 100(2):490–509, 2018.
- P. L. Berger and T. Luckmann. *The social construction of reality: A treatise in the sociology of knowledge*. Doubleday, Garden City, NY, 1966.
- A. Bernstein, Y. Hadash, Y. Lichtash, G. Tanay, K. Shepherd, and D. M. Fresco. Decentering and related constructs: A critical review and metacognitive processes model. *Perspectives on Psychological Science*, 10(5):599–617, 2015. doi: 10.1177/1745691615594577.
- E. Bernstein, J. Shore, and D. Lazer. How intermittent breaks in interaction improve collective intelligence. *Proceedings of the National Academy of Sciences*, 115(35):8734–8739, 2018.
- A. Birhane, W. Isaac, V. Prabhakaran, M. Diaz, M. C. Elish, I. Gabriel, and S. Mohamed. Power to the people? opportunities and challenges for participatory ai. In *Proceedings of the 2nd ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*, pages 1–8, 2022.
- R. Bommasani. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.
- N. Bostrom. *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press, 2014. ISBN 0199678111.
- P. Brown. A rhythmic mechanism for communication in the cortex. *Trends in neurosciences*, 26(5):232–233, 2003.
- E. Brynjolfsson, D. Li, and L. Raymond. Generative ai at work. *The Quarterly Journal of Economics*, 140(2):889–942, 2025.
- G. Canguilhem. *Le normal et le pathologique*. Presses Universitaires de France, Paris, 1966.
- A. J. Chaney, B. M. Stewart, and B. E. Engelhardt. How algorithmic confounding in recommendation systems increases homogeneity and decreases utility. In *Proceedings of the 12th ACM conference on recommender systems*, pages 224–232, 2018.

- D. Charlesworth, X. Vekemans, V. Castric, and S. Glémin. Plant self-incompatibility systems: a molecular evolutionary perspective. *New Phytologist*, 168(1):61–69, 2005.
- A. Chatterji, T. Cunningham, D. Deming, Z. Hitzig, C. Ong, C. Shan, and K. Wadman. How people use ChatGPT. Technical report, OpenAI, Sept. 2025. URL <https://cdn.openai.com/pdf/a253471f-8260-40c6-a2cc-aa93fe9f142e/economic-research-chatgpt-usage-paper.pdf>.
- D. R. Chialvo. Emergent complex neural dynamics. *Nature physics*, 6(10):744–750, 2010.
- N. Chomsky. *Syntactic Structures*. Mouton de Gruyter, The Hague, 1957.
- L. Cocchi, L. L. Gollo, A. Zalesky, and M. Breakspear. Criticality in the brain: A synthesis of neurobiology, models and cognition. *Progress in neurobiology*, 158:132–152, 2017.
- T. H. Costello, G. Pennycook, and D. G. Rand. Durably reducing conspiracy beliefs through dialogues with ai. *Science*, 2024.
- P. Dayan, Y. Niv, B. Seymour, and N. D. Daw. The misbehavior of value and the discipline of the will. *Neural networks*, 19(8):1153–1160, 2006.
- G. Deco, V. K. Jirsa, and A. R. McIntosh. Emerging concepts for the dynamical organization of resting-state activity in the brain. *Nature Reviews Neuroscience*, 12(1):43–56, 2011.
- S. Dehaene, F. Al Roumi, Y. Lakretz, S. Planton, and M. Sablé-Meyer. Symbols and mental programs: a hypothesis about human singularity. *Trends in Cognitive Sciences*, 26(9):751–766, 2022.
- J. Dewey. Theory of valuation. *International encyclopedia of unified science*, 1939.
- A. Diamond. Executive functions. *Annual review of psychology*, 64(1):135–168, 2013.
- H. K. Dooner and I. M. Martínez-Férez. Recombination occurs uniformly within the bronze gene, a meiotic recombination hotspot in the maize genome. *The Plant Cell*, 9(9):1633–1646, 1997. doi: 10.1105/tpc.9.9.1633.
- A. R. Doshi and O. P. Hauser. Generative ai enhances individual creativity but reduces the collective diversity of novel content. *Science advances*, 10(28):eadn5290, 2024.
- K. Duncker. On problem-solving. *Psychological Monographs*, 58, 1945.
- European Parliament and Council of the European Union. Regulation (eu) 2024/1689 of the european parliament and of the council of 13 june 2024 laying down harmonised rules on artificial intelligence. Official Journal of the European Union L, 2024. URL <https://eur-lex.europa.eu/eli/reg/2024/1689/oj>.
- J. Felsenstein. The evolutionary advantage of recombination. *Genetics*, 78(2):737–756, 1974. doi: 10.1093/genetics/78.2.737.
- R. A. Fisher. *The Genetical Theory of Natural Selection*. The Clarendon Press, Oxford, 1930.
- J. A. Fodor. *The Language of Thought*. Harvard University Press, Cambridge, MA, 1975.
- V. Frey and A. Van de Rijt. Social influence undermines the wisdom of the crowd in sequential decision making. *Management science*, 67(7):4273–4286, 2021.
- G. O. Gabbard and E. P. Lester. *Boundaries and boundary violations in psychoanalysis*. American Psychiatric Publishing, 1995.
- I. Gabriel and G. Keeling. A matter of principle? ai alignment as the fair treatment of claims. *Philosophical Studies*, 182(7):1951–1973, 2025. doi: 10.1007/s11098-025-02300-4.
- E. Goffman. *Frame analysis: An essay on the organization of experience*. Harvard university press, 1974.
- C. L. Grady and D. D. Garrett. Understanding variability in the bold signal and why it matters for aging. *Brain Imaging and Behavior*, 8:274–282, 2014. doi: 10.1007/s11682--013-9253--0.
- D. Hadfield-Menell, A. D. Dragan, P. Fisac, and S. Russell. Cooperative inverse reinforcement learning. In D. D. Lee, M. Sugiyama, U. V. Luxburg, I. Guyon, and R. Garnett, editors, *Advances in Neural Information Processing Systems 29 (NIPS 2016)*, pages 3909–3917, 2016.
- D. Hadfield-Menell, A. D. Dragan, and S. Russell. The off-switch game. In *Proceedings of the 31st AAAI Conference on Artificial Intelligence (AAAI 2017)*, pages 220–227, 2017.
- J. Haidt. *The righteous mind: Why good people are divided by politics and religion*. Vintage, 2012.
- C. Haldeman and J. M. Beggs. Critical branching captures activity in living neural networks and maximizes the number of metastable states. *Physical Review Letters*, 94(5):058101, 2005. doi: 10.1103/PhysRevLett.94.058101.
- C. Hammond, H. Bergman, and P. Brown. Pathological synchronization in parkinson’s disease: networks, models and treatments. *Trends in neurosciences*, 30(7):357–364, 2007.
- W. A. Hershberger. An approach through the looking-glass. *Animal Learning & Behavior*, 14(4):443–451, 1986.
- L. Hong and S. E. Page. Groups of diverse problem solvers can outperform groups of high-ability problem solvers. *Proceedings of the National Academy of Sciences*, 101(46):16385–16389, 2004.
- E. Hutchins. *Cognition in the Wild*. MIT Press, Cambridge, MA, 1995.
- J. S. Isaacson and M. Scanziani. How inhibition shapes cortical activity. *Neuron*, 72(2):231–243, 2011.
- I. L. Janis. *Victims of groupthink: A psychological study of foreign-policy decisions and fiascoes*. Houghton Mifflin, 1972.
- H. J. Jeon, S. Milli, and A. Dragan. Reward-rational (implicit) choice: A unifying formalism for reward learning. *Advances in Neural Information Processing Systems*, 33:4415–4426, 2020.
- Y. J. John, L. Caldwell, D. E. McCoy, and O. Braganza. Dead rats, dopamine, performance metrics, and peacock tails: Proxy failure is an inherent risk in goal-oriented systems. *Behavioral and Brain Sciences*, pages 1–68, 2023.
- S. Johnson. *Where Good Ideas Come From: The Natural History of Innovation*. Riverhead Books, New York, 2010.
- T. B. Kashdan and J. Rottenberg. Psychological flexibility as a fundamental aspect of health. *Clinical psychology review*, 30(7):865–878, 2010.
- S. A. Kauffman. The origins of order: Self-organization and selection in evolution. In *Spin glasses and biology*, pages 61–100. World Scientific, 1992.
- R. Kegan. *The evolving self: Problem and process in human development*. Harvard University Press, 1982.
- V. Krakovna, A. Gleave, and J. Miller. Specification gaming: The flip side of AI ingenuity. DeepMind Safety Research Blog, May 2020. URL <https://www.deepmind.com/blog/specification-gaming-the-flip-side-of-ai-ingenuity>. Accessed on 2025-10-09.
- Z. Kurth-Nelson, T. Behrens, G. Wayne, K. Miller, L. Luettgau, R. Dolan, Y. Liu, and P. Schwartenbeck. Replay and compositional computation. *Neuron*, 111(4):454–469, 2023.
- B. M. Lake, R. Salakhutdinov, and J. B. Tenenbaum. Human-level concept learning through probabilistic program induction. *Science*, 350(6266):1332–1338, 2015. doi: 10.1126/science.aab3050.
- C. G. Langton. Computation at the edge of chaos: Phase transitions and emergent computation. *Physica D: nonlinear phenomena*, 42(1-3):12–37, 1990.
- B. Latour. *We have never been modern*. Harvard university press, 2012.
- D. Lazer and A. Friedman. The network structure of exploration and exploitation. *Administrative Science Quarterly*, 52(4):667–694, 2007.
- J. Z. Leibo, A. S. Vezhnevets, W. A. Cunningham, S. Krier, M. Diaz, and S. Osindero. Societal and technological progress as sewing an ever-growing, ever-changing, patchy, and polychrome quilt. *arXiv preprint arXiv:2505.05197*, 2025.
- J. Z. Leibo, A. S. Vezhnevets, M. Diaz, J. P. Agapiou, W. A. Cunningham, P. Sunehag, L. Cross, R. Koster, S. M. Bileschi, M. Chang, et al. A theory of appropriateness that accounts for norms of rationality. *arXiv preprint arXiv:2603.14050*, 2026.
- T. Lenormand. Gene flow and the limits to natural selection. *Trends in Ecology & Evolution*, 17(4):183–189, 2002. doi: 10.1016/S0169-5347(02)02497-7.
- J. K. Leutgeb, S. Leutgeb, M.-B. Moser, and E. I. Moser. Pattern separation in the dentate gyrus and ca3 of the hippocampus. *science*, 315(5814):961–966, 2007.

- A. Livnat, C. Papadimitriou, J. Dushoff, and M. W. Feldman. A mixability theory for the role of sex in evolution. *Proceedings of the National Academy of Sciences*, 105(50):19803–19808, 2008.
- A. Livnat, C. Papadimitriou, N. Pippenger, and M. W. Feldman. Sex, mixability, and modularity. *Proceedings of the National Academy of Sciences*, 107(4):1452–1457, 2010.
- J. Lorenz, H. Rauhut, F. Schweitzer, and D. Helbing. How social influence can undermine the wisdom of crowd effect. *Proceedings of the national academy of sciences*, 108(22):9020–9025, 2011.
- A. Lutz, A. P. Jha, J. D. Dunne, and C. D. Saron. Investigating the phenomenological matrix of mindfulness-related practices from a neurocognitive perspective. *American Psychologist*, 70(7):632–658, 2015. doi: 10.1037/a0039585.
- D. A. Marques, J. I. Meier, and O. Seehausen. A combinatorial view on speciation and adaptive radiation. *Trends in ecology & evolution*, 34(6):531–544, 2019.
- H. R. Maturana and F. J. Varela. *Autopoiesis and cognition: The realization of the living*. Springer Science & Business Media, 2012.
- J. Maynard Smith. The origin and maintenance of sex. In *Group selection*, pages 163–175. Aldine Atherton, 1971.
- D. J. McFarland and R. M. Sibly. The behavioural final common path. *Philosophical Transactions of the Royal Society of London B*, 270(907):265–293, 1975.
- L. Messeri and M. J. Crockett. Artificial intelligence and illusions of understanding in scientific research. *Nature*, 627(8002):49–58, 2024.
- E. K. Miller and J. D. Cohen. An integrative theory of prefrontal cortex function. *Annual review of neuroscience*, 24(1):167–202, 2001.
- S. Minuchin. *Families and Family Therapy*. Harvard University Press, 1974.
- H. J. Muller. Some genetic aspects of sex. *The American Naturalist*, 66(703):118–138, 1932.
- H. J. Muller. The relation of recombination to mutational advance. *Mutation Research*, 1(1):2–9, 1964.
- P. M. Müller, G. Miron, M. Holtkamp, and C. Meisel. Critical dynamics predicts cognitive performance and provides a common framework for heterogeneous mechanisms impacting cognition. *Proceedings of the National Academy of Sciences*, 122(14):e2417117122, 2025.
- T. Nagel. The fragmentation of value. In *Mortal Questions*, pages 128–141. Cambridge University Press, Cambridge, 1979.
- T. Nagel. *The View from Nowhere*. Oxford University Press, New York, 1986. ISBN 9780195056440.
- D. C. North. *Institutions, Institutional Change and Economic Performance*. Cambridge University Press, Cambridge, 1990.
- M. C. Nussbaum. *The fragility of goodness: Luck and ethics in Greek tragedy and philosophy*. Cambridge University Press, 2001.
- S. Ohlsson. Information-processing explanations of insight and related phenomena. *Advances in the psychology of thinking*, 1: 1–44, 1992.
- C. O’Keefe, P. Cihon, B. Garfinkel, C. Flynn, J. Leung, and A. Dafoe. The windfall clause: Distributing the benefits of ai for the common good. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, pages 327–331, 2020.
- C. Olah, N. Cammarata, L. Schubert, G. Goh, M. Petrov, and S. Carter. Zoom in: An introduction to circuits. *Distill*, 5(3):e00024–001, 2020.
- OpenAI. Pacing model development in an era of cyber-critical capabilities, 2026. URL <https://openai.com/index/pacing-model-development-cyber-capabilities/>. Accessed: 2026-08-21.
- T. Ord. *The precipice: Existential risk and the future of humanity*. Hachette UK, 2020.
- E. Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, Cambridge, 1990.
- S. P. Otto. The evolutionary enigma of sex. *The American Naturalist*, 174(S1):S1–S14, 2009. doi: 10.1086/599084.
- V. Padmakumar and H. He. Does writing with language models reduce content diversity? *arXiv preprint arXiv:2309.05196*, 2023.
- S. E. Page. *The difference: How the power of diversity creates better groups, firms, schools, and societies*. Princeton University Press, 2008.
- K. Paigen and P. Petkov. Mammalian recombination hot spots: Properties, control and evolution. *Nature Reviews Genetics*, 11(3):221–233, 2010. doi: 10.1038/nrg2712.
- R. T. Paine. Food web complexity and species diversity. *The American Naturalist*, 100(910):65–75, 1966. doi: 10.1086/282400.
- F. Perls, R. Hefferline, and P. Goodman. *Gestalt Therapy: Excitement and Growth in the Human Personality*. Julian Press, 1951.
- S. Pinker. *The Language Instinct: How the Mind Creates Language*. William Morrow and Company, New York, 1994.
- M. Polanyi. *The Tacit Dimension*. Doubleday, Garden City, NY, 1966.
- I. Prigogine and I. Stengers. *Order Out of Chaos: Man’s New Dialogue with Nature*. Bantam Books, New York, 1984.
- M. I. Rabinovich, R. Huerta, P. Varona, and V. S. Afraimovich. Transient cognitive dynamics, metastability, and decision making. *PLoS Computational Biology*, 4(5):e1000072, 2008. doi: 10.1371/journal.pcbi.1000072.
- R. P. N. Rao and D. H. Ballard. Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2(1):79–87, 1999. doi: 10.1038/4580.
- J. Rawls. *Political Liberalism*. Columbia University Press, New York, 1993. ISBN 9780231052481.
- N. Rosenberg. The direction of technological change: inducement mechanisms and focusing devices. *Economic Development and Cultural Change*, 18(1, Part 1):1–24, 1969.
- S. Russell. *Human Compatible: Artificial Intelligence and the Problem of Control*. Penguin Publishing Group, 2019. ISBN 9780525558620. URL <https://books.google.co.uk/books?id=M1eFDwAAQBAJ>.
- L. Sagan. On the origin of mitosing cells. *Journal of Theoretical Biology*, 14(3):225–274, 1967. doi: 10.1016/0022-5193(67)90079-3.
- D. L. Schacter, D. R. Addis, and R. L. Buckner. Remembering the past to imagine the future: the prospective brain. *Nature reviews neuroscience*, 8(9):657–661, 2007.
- P. Scharre. *Army of None: Autonomous Weapons and the Future of War*. W. W. Norton, New York, 2018.
- E. Schrödinger. *What is Life? The Physical Aspect of the Living Cell*. Cambridge University Press, Cambridge, UK, 1944. Based on lectures delivered at Trinity College, Dublin, February 1943.
- R. Shah, P. Freire, N. Alex, R. Freedman, D. Krashennikov, L. Chan, M. D. Dennis, P. Abbeel, A. Dragan, and S. Russell. Benefits of assistance over reward learning. *NeurIPS*, 2020.
- W. L. Shew, H. Yang, S. Yu, R. Roy, and D. Plenz. Information capacity and transmission are maximized in balanced cortical networks with neuronal avalanches. *Journal of Neuroscience*, 31(1):55–63, 2011. doi: 10.1523/JNEUROSCI.4637--10.2011.
- J. Sidanius and F. Pratto. *Social dominance: An intergroup theory of social hierarchy and oppression*. Cambridge University Press, 2001.
- P. Smolensky. Tensor product variable binding and the representation of symbolic structures in connectionist systems. *Artificial Intelligence*, 46(1–2):159–216, 1990. doi: 10.1016/0004-3702(90)90007-M.
- T. Sorensen, J. Moore, J. Fisher, M. Gordon, N. Miresghallah, C. M. Rytting, A. Ye, L. Jiang, X. Lu, N. Dziri, et al. A roadmap to pluralistic alignment. *arXiv preprint arXiv:2402.05070*, 2024.
- Z. Sourati, A. S. Ziabari, and M. Dehghani. The homogenizing effect of large language models on human expression and thought. *Trends in Cognitive Sciences*, 2026.
- D. Speijer, J. Lukeš, and M. Eliáš. Sex is a ubiquitous, ancient, and inherent attribute of eukaryotic life. *Proceedings of the National Academy of Sciences*, 112(29):8827–8834, 2015.
- K. O. Stanley, J. Lehman, and L. B. Soros. Open-endedness: The last grand challenge you’ve never heard of. O’Reilly Radar, 2017. URL <https://www.oreilly.com/radar/open-endedness-the-last-grand-challenge-youve-never-heard-of/>. Accessed: 2025-01-28.

- P. D. Stokes. *Creativity from Constraint: The Psychology of Breakthrough*. Springer, New York, 2005.
- J. Surowiecki. *The wisdom of crowds*. Vintage, 2005.
- D. Susskind. *A World Without Work: Technology, Automation, and How We Should Respond*. Metropolitan Books, New York, 2020.
- E. Tognoli and J. S. Kelso. The metastable brain. *Neuron*, 81(1): 35–48, 2014.
- D. S. Touretzky and G. E. Hinton. A distributed connectionist production system. *Cognitive Science*, 12(3):423–466, 1988. doi: 10.1207/s15516709cog1203_3.
- G. P. Wagner and L. Altenberg. Perspective: complex adaptations and the evolution of evolvability. *Evolution*, 50(3):967–976, 1996.
- R. A. Watson and J. B. Pollack. A computational model of symbiotic composition in evolutionary transitions. *BioSystems*, 69 (2–3):187–209, 2003. doi: 10.1016/S0303-2647(02)00135-1.
- R. A. Watson and E. Szathmáry. How can evolution learn? *Trends in ecology & evolution*, 31(2):147–157, 2016.
- G. C. Williams. *Sex and Evolution*. Princeton University Press, Princeton, NJ, 1975.
- L. Wittgenstein. *Philosophical Investigations*. Blackwell, Oxford, 1953.
- A. W. Woolley, C. F. Chabris, A. Pentland, N. Hashmi, and T. W. Malone. Evidence for a collective intelligence factor in the performance of human groups. *Science*, 330(6004):686–688, 2010.
- J. P. Wright, C. G. Jones, and A. S. Flecker. An ecosystem engineer, the beaver, increases species richness at the landscape scale. *Oecologia*, 132(1):96–101, 2002. doi: 10.1007/s00442-002-0929-1.
- M. A. Yassa and C. E. Stark. Pattern separation in the hippocampus. *Trends in neurosciences*, 34(10):515–525, 2011.
- E. Yudkowsky. Coherent extrapolated volition. *Singularity Institute for Artificial Intelligence*, 2004.
- V. Zimorski, C. Ku, W. F. Martin, and S. B. Gould. Endosymbiotic theory for organelle origins. *Current Opinion in Microbiology*, 22:38–48, 2014. doi: 10.1016/j.mib.2014.09.008.
- K. J. S. Zollman. The epistemic benefit of transient diversity. *Erkenntnis*, 72(1):17–35, 2010.